

COMMUNICATIONS TEST 2

Number of Questions: 35

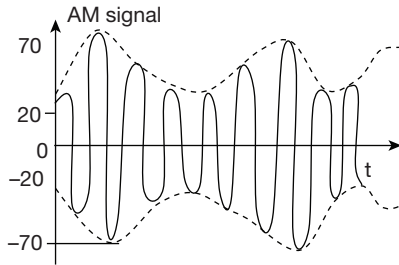
Time: 60 min.

Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. A PCM system uses a uniform quantizer followed by a 8 bit encoder. The bit rate is 64 kbps. The B.W of message signal for which the system works satisfactorily is

(A) 1 kHz (B) 2 kHz
(C) 4 kHz (D) 8 kHz

2. Percentage modulation of the signal shown in figure is

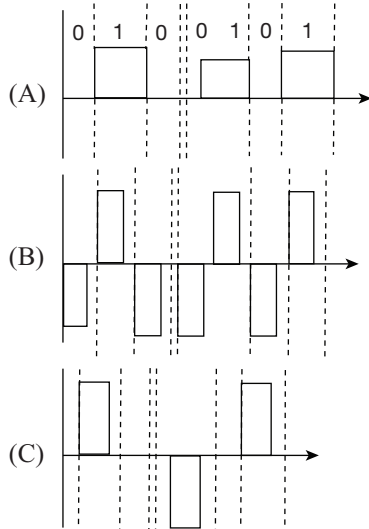


(A) 14.9 (B) 28.5
(C) 11.8 (D) 55.6

3. Power spectral density of a random process is

(A) an even function of frequency ω
(B) a non negative function of ω
(C) is a real function of frequency ω
(D) All of the above

4. For the binary sequence 0100101. Bipolar RZ signaling format is



(D) None of these

5. If a binary PCM if 1 occur with probability $\frac{3}{4}$ and '0' occur with probability equal to $\frac{1}{4}$, then calculate

amount of information carries by bit '1' and '0' are

(A) 0.415 bit and 2 bit (B) 2 bits and 0.415 bits
(C) 1 bit and 1 bit (D) 2 bit and 1 bit

6. Which of the following is not a component of phase locked loop?

(A) VCO (B) Phase detector
(C) Frequency multiplier (D) Loop filter

7. The differential entropy $h(x)$ of the uniformly distributed random variable x with the following probability

$$\text{density function } f_x(x) = \begin{cases} \frac{1}{b} & \text{for } 0 \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad \text{for } b = 1$$

is _____

(A) zero (B) 1
(C) $\frac{1}{2}$ (D) -1

8. Consider the following statement comparing DM and PCM systems. DM requires

(1) a lower sampling rate
(2) a higher sampling rate
(3) Band width depends on slope of the signal
(4) simple hardware

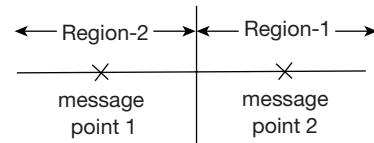
The correct statements are

(A) 1, 3 and 4 (B) 1, 2 and 3
(C) 2, 3 and 4 (D) 1, 2 and 4

9. Trans mux is an equipment used to convert

(A) FDM in to TDM (B) TDM in to FDM
(C) Both (A) and (B) (D) None of the above

10. The signal space diagram given below



Corresponds to

(A) BFSK (B) BPSK
(C) BASK (D) QPSK

11. Power saving in DSBSC system is, when modulation index is unity

(A) 67% (B) 78%
(C) 100% (D) 33%

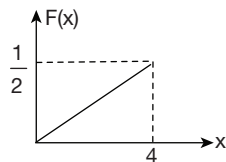
12. If the amplitude of message signal $m(t)$ is doubled, then

(A) bandwidth of PM doubles but bandwidth of FM remains same
(B) bandwidth of FM doubles but bandwidth of PM remains same
(C) bandwidth of both doubles
(D) bandwidth of both remains same

13. The output of the matched filter is proportional to
 (A) input signal itself
 (B) shifted version of autocorrelation function of the input signal
 (C) auto correlation function of the input signal
 (D) None of these
14. Zero ISI can be achieved if the impulse response of channel is
 (A) Rectangular pulse (B) Triangular pulse
 (C) sinc pulse (D) $\sin c^2$ pulse
15. Near far problem can be reduced in CDMA by
 (A) increasing the power of "far" users and decreasing the power of "near" user
 (B) decreasing the power of "far" users and increasing the power of "near" users
 (C) increasing the power of "far" and "near" users
 (D) decreasing the power of "far" and "near" users
16. For a certain binary non symmetric channel it is given that $P_{y/x}(0/1) = 0.1$ and $P_{y/x}(1/0) = 0.2$. Where x is transmitted digit and y is received digit. If $P_x(0) = 0.4$, then $P_y(0)$ and $P_y(1)$ are
 (A) 0.44 and 0.58 (B) 0.72 and 0.42
 (C) 0.42 and 0.56 (D) 0.58 and 0.44
17. A signal $m(t)$ of bandwidth $B = 4$ KHz is transmitted using a binary companded PCM with $\mu = 100$. The output SNR and transmission bandwidth for quantization levels $L = 256$ and $L = 64$ are _____ for $L = 256$ _____
 (A) SNR is 16 times greater and 33% more B.W required
 (B) SNR is 12 times greater and 25% more B.W required
 (C) SNR is 16 times greater and 25% more B.W required
 (D) None of these
18. An angle modulated signal with carrier frequency $\omega_c = 2\pi \times 10^5$ rad/sec is described by the

$$\varphi(t) = 10 \cos [\omega_c t + 5 \sin 3000 t + 10 \sin 2000\pi t].$$

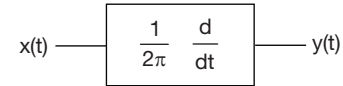
 The deviation ratio β is
 (A) 12.393 (B) 77.831
 (C) 13.393 (D) 35.000
19. The input signal to a quantizer have a PDF as shown in figure



The signal power at quantizer output is

- (A) 2 (B) 4
 (C) 8 (D) 10

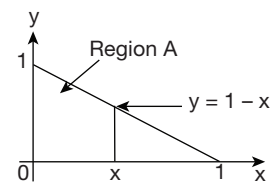
20. An Analog signal $x(t) = 5 \cos(4\pi 10^3 t)$ is sampled at nyquist rate and quantized into nyquist rate and quantized in to 1024 levels. The time duration T of 1 bit of binary encoded signal is
 (A) 5 μ sec (B) 50 μ sec
 (C) 25 μ sec (D) 2.5 μ sec
21. A deterministic signal $x(t) = \sin 2\pi t$ is passed through a differentiator as shown in figure



The cross correlation function $R_{xy}(0)$ at $\tau = 0$ is

- (A) 0 (B) π
 (C) $\frac{\pi}{2}$ (D) 2

22. The joint probability density function of two random variables x and y is given by $f_{xy}(x,y) = c(1-x-y)$ for values of x and y for which (x,y) lies within the triangle as shown in figure



The value of constant C is

- (A) 2 (B) 4
 (C) 6 (D) 8

23. Match the following List I with List II

List I

List II

- | | |
|--------------------|--|
| (1) ASK | (i) $P_e = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{E}{N_o}}$ |
| (2) PSK | (ii) $P_2 = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{0.6E}{N_o}}$ |
| (3) FSK | (iii) $P_e = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{E}{4N_o}}$ |
| (4) Matched filter | (iv) $P_e = \frac{1}{2} \operatorname{erfc} \sqrt{\frac{2E}{N_o}}$ |
- (A) 1 – (iii) 2 – (i) 3 – (ii) 4 – (iv)
 (B) 1 – (iii) 2 – (i) 3 – (ii) 4 – (iv)
 (C) 1 – (iii) 2 – (iv) 3 – (ii) 4 – (i)
 (D) 1 – (ii) 2 – (iii) 3 – (iv) 4 – (i)

24. Eight signals of 2 kHz BW each using PCM technique and 8 bits are used to encode one sample. If the sampling rate is 25% above the nyquist rate of raised cosine pulses are used with roll off factor $\alpha = 0.2$. Then the minimum transmission bandwidth required is
 (A) 128 kHz (B) 153.6 kHz
 (C) 160 kHz (D) 192 kHz

25. A Source delivers symbols m_1, m_2, m_3 and m_4 , with probabilities $\frac{1}{4}, \frac{1}{4}, \frac{1}{8}$ and $\frac{3}{8}$ respectively.

The entropy of the system is

- (A) 1.905 bits per symbol
(B) 1.9 bits per second
(C) 1.7 symbols per second
(D) 1.85 symbols per bit

26. Let a message signal $m(t)$ be transmitted using single sideband modulation

The power spectral density of $m(t)$ is

$$S_m(f) = \begin{cases} \frac{a|f|}{W} & \text{for } |f| \leq W \\ 0 & \text{elsewhere} \end{cases}$$

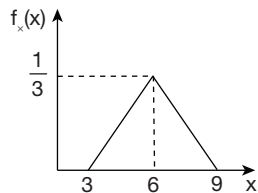
A white Gaussian noise of zero mean and power spectral density $\frac{N_o}{2}$ is added to the SSB modulated wave at the receiver input. Derive an expression for the output signal to noise ratio of the receiver

- (A) $\frac{aA_c}{2N_o}$ (B) $\frac{aA_c^2}{4N_o}$
(C) $\frac{a}{2N_o}$ (D) $\frac{A_c}{2N_o}$

27. Polar binary pulses are received with peak amplitude $A_p = 1$ mV. The channel noise rms amplitude is $235.8 \mu V$. Threshold detection is used and 1 and 0 are equally likely. The detection error probability is if $Q(x)$ is given as

- 2.1 1.786×10^{-2}
3.68 1.166×10^{-4}
4.24 1.118×10^{-5}
5.2 9.964×10^{-8}
(A) 1.786×10^{-2} (B) 1.166×10^{-4}
(C) 1.118×10^{-5} (D) 9.964×10^{-8}

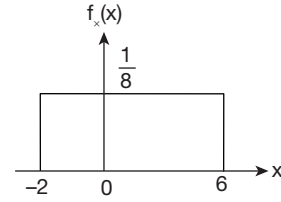
28. The PDF of a random variable x is shown below



The probability of the event $p(x = 4.5)$ is _____

- (A) $\frac{1}{6}$ (B) $\frac{1}{3}$
(C) $\frac{1}{9}$ (D) Zero

29. For a random variable x following the probability density function, $p(x)$ is shown in figure. The mean and variance are respectively

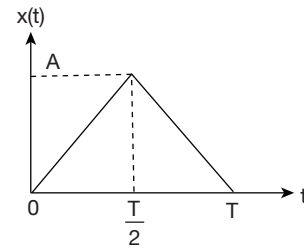


- (A) 2, 5.33 (B) 4, 18.66
(C) 3, 14.66 (D) 3, 18.66

30. In a CCIT hierarchy, first level multiplexer uses 30 voice signals by using PCM signals. The output bit rate of this multiplexer is 2.304 Mbps. If bit robbing is not implemented, then the number of framing plus signaling bits/frame are _____

- (A) 8 (B) 14
(C) 16 (D) 18

31. The maximum signal to noise ratio of the matched filter for a triangular pulse given in figure is noise on the channel is white noise and is given as $\frac{N_o}{2}$



- (A) $\frac{A^2 T}{6N_o}$ (B) $\frac{2A^2 T}{3N_o}$
(C) $\frac{A^2 T}{3N_o}$ (D) $\frac{2A^2 T}{5N_o}$

Common data for Questions 32 and 33:

A communication system having following characteristics

$$S_x = E[x^2(t)] = \frac{1}{2}$$

$$B = 10 \text{ kHz}$$

$$\frac{\eta}{2} = 10^{-12} \text{ W/Hz}$$

Transmission loss = 70 dB

32. The required transmission power S_T needed to achieve $\frac{S}{N_o} = 40$ dB, when the modulation is SSB

- (A) 1 KW (B) 2 KW
(C) 1 W (D) 2 W

33. The transmission bandwidth of FM signal is used with deviation ratio $D = 5$

- (A) 110 kHz (B) 120 kHz
(C) 100 kHz (D) 20 kHz

Statement for Linked answer Questions 34 and 35:

In a binary PCM system, the output signal to quantizing noise ratio is to be held to a minimum value of 30 dB

34. Number of required levels are

- (A) 25 (B) 26
(C) 31 (D) 32

35. The corresponding output signal to quantizing noise ratio is

- (A) 25.84 dB (B) 31.86 dB
(C) 30.86 dB (D) 26.84 dB

ANSWER KEYS

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. D | 3. D | 4. B | 5. A | 6. C | 7. A | 8. C | 9. C | 10. A |
| 11. A | 12. C | 13. B | 14. C | 15. A | 16. C | 17. A | 18. A | 19. C | 20. C |
| 21. A | 22. C | 23. C | 24. D | 25. A | 26. B | 27. C | 28. A | 29. A | 30. D |
| 31. B | 32. B | 33. B | 34. D | 35. B | | | | | |

HINTS AND EXPLANATIONS

1. Sampling frequency $f_s = \frac{64}{8}$ kbps = 8 KHz

So message signal could have 4kHz. B.W.

Choice (C)

2. $\mu_a = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{70 - 20}{70 + 20} = \frac{50}{90} = 0.556$

% modulation = $0.556 \times 100 = 55.6\%$

Choice (D)

3. All three are proportion of PSD of a random process.

Choice (D)

4. In Bipolar '0' is represented as negative value and 'one' is represented as positive value and In RZ (Return to zero) format, in between the clock duration pulse return to zero.

Choice (B)

5. Binit '0' has $P(x_0) = \frac{1}{4}$ and binit '1' has $P(x_1) = \frac{3}{4}$

Then, amount of information given as

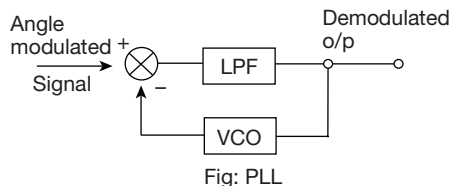
$$I(x_i) = \log_2 \frac{1}{p(x_i)}$$

$$I(x_0) = \log_2 4 = 2 \text{ bits and } P(x_1) = \frac{3}{4}$$

$$I(x_2) = \log_2 \left(\frac{4}{3} \right) = 0.415 \text{ bits.}$$

Choice (A)

6.



Choice (C)

7. We know the differential entropy x is given by

$$H(x) = - \int_{-\infty}^{+\infty} f_x(x) \log_2 f_x(x) dx$$

$$H(x) = - \int_0^b \frac{1}{b} \log_2 \frac{1}{b} dx = \log_2 b$$

For $b = 1$

$$H(x) = \log_2 1 = 0.$$

Choice (A)

8. DM system have higher sampling rate than PCM system, and requires simple hardware to construct.

Choice (C)

9. Trans mux is used to convert both TDM to FDM and FDM to TDM.

Choice (C)

10. Choice (A)

11. In AM system when modulation index is 1 then the total 33% of power is utilized, and rest 67% of total power get wasted.

But in DSB SC total 100% power is utilized so 67% power get saved.

Choice (A)

12. For FM $S(t) = A_c \cos(\omega_c t + 2\pi k_f \int m(t) dt)$

So phase $\theta_{iFM} = \omega_c t + k_f \int m(t) dt$

$$\omega_{iFM} = \omega_c + K_f m(t)$$

Now if amplitude of $m(t)$ gets doubled then $\omega_{iFM} = \omega_c + 2K_f m(t)$

$$\Delta \omega_{FM} = |2K_f m(t)|_{\max}$$

Means FM doubles its frequency deviation, so bandwidth also

Now for PM

$$\phi_{PM}(t) = A \cos(\omega_c t + K_p m(t))$$

So phase $\theta_{iPM} = \omega_c t + K_p m(t)$

$$\omega_{iPM} = \omega_c + K_p \frac{dm(t)}{dt}$$

So amplitude of $m(t)$ is doubled

$$\text{Then new } \omega_{iPM}^1 = \omega_c + 2K_p \frac{dm(t)}{dt}$$

$$\Delta \omega_{PM} = 2K_p \frac{dm(t)}{dt}$$

So deviation for PM also gets doubled.

So bandwidth also.

Choice (C)

13. Output of matched filter $x_0(t) = \frac{2K}{N_0} R(t - T)$

Output of matched filter is proportional to autocorrelation function of $x(t)$ shifted to T . where $\frac{2K}{N_0}$ is constant term and it depends upon psd of noise. Choice (B)

14. ISI can be reduced to zero by using sinc pulse.

Choice (C)

15. In CDMA near far problem is reduced by increasing the power of far users and by decreasing the power of "near" users it is also called power control in CDMA.

Choice (A)

16. $P_{y/x}(0/1) = 0.1$

$$P_{y/x}(1/0) = 0.2$$

$$P_x(0) = 0.4$$

$$P_x(1) = 0.6$$

$$P_{y/x}(0/0) = 1 - 0.1 = 0.9$$

$$P_{y/x}(1/1) = 1 - 0.2 = 0.8$$

$$P_y(1) = P_x(0) P_{y/x}(1/0) + P_x(1) P_{y/x}(1/1)$$

$$= 0.4 \times 0.2 + 0.6 \times 0.8$$

$$= 0.08 + 0.48 = 0.56$$

$$P_y(0) = P_x(1) P_{y/x}(0/1) + P_x(0) P_{y/x}(0/0)$$

$$= 0.6 \times 0.1 + 0.4 \times 0.9$$

$$= 0.06 + 0.36 = 0.42.$$

Choice (C)

17. For $L = 64$, no of bits required = 6

So transmission bandwidth = nB

Out put SNR = $(\alpha + 6n) dB$

$$\alpha = 10 \log \frac{3}{[\ln(1 + \mu)]^2} = -8.51$$

SNR = 27.49 dB

For $L = 256$, no of bits required $n = 8$

So transmission bandwidth = $8 \times 4 = 32$ KHz

Output SNR = $\alpha + 6n = 39.49$ dB

Difference between two SNR = 12 dB

Or ratio is 15.84 or 16% greater for $L = 256$

Difference between two bandwidth are 8 KHz

So % of more bandwidth required for $L = 256$ is

$$= \frac{8}{24} \times 100 = 33\%.$$

Choice (A)

18. Signal bandwidth is highest frequency in $m(t)$ So

$$B = \frac{2000\pi}{2\pi} = 1000 \text{ Hz or } 1 \text{ KHz}$$

$$\text{Or } \omega_i = \frac{d}{dt} \theta_i(t)$$

$$= \omega_c + 15000 \cos 3000 t + 20000 \pi \cos 2000 \pi t$$

$$\Delta \omega = |15000 \cos 3000 t + 20000 \pi \cos 2000 \pi t|_{\max}$$

$$\Delta \omega = 77831.85$$

$$\Delta f = \frac{\Delta \omega}{2\pi}$$

$$\Delta f = 12393.60 \text{ Hz}$$

$$\text{Deviation ratio } \beta = \frac{\Delta f}{B} = 12.393.$$

Choice (A)

$$19. \text{ Here } P(x) = \begin{cases} \frac{x}{8}, & 0 \leq x \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

$$\text{So signal power } E[x^2] = \int_0^4 p(x) x^2 dx$$

$$= \frac{1}{8} \int_0^4 x^3 dx = 8.$$

Choice (C)

20. Total number of equalization levels are 1024

So number of bits used to encode the signal is $N = \log_2 1024 = 10$ bits

$x(t)$ have frequency component $\omega_m = 4\pi 10^3$

$$f_m = 2 \times 10^3 = 2 \text{ KHz}$$

So Nyquist rate $f_s = 4 \text{ Hz}$

Now $N f_s$ binary pulses must be transmitted per second

$$\text{So } T = \frac{1}{N f_s} = \frac{1}{10 \times 4 \times 10^3}$$

$$= \frac{10^{-4}}{4} = 25 \mu \text{ sec.}$$

Choice (C)

21. $x(t) = \sin(2\pi t)$

$$y(t) = \frac{1}{2\pi} \frac{d}{dt} x(t)$$

$$= \frac{1}{2\pi} \cdot 2\pi \cdot \cos 2\pi t = \cos 2\pi t$$

$$y(t) = \cos(2\pi t)$$

$$\text{Now, } R_{xy}(\tau) = E[x(t) \cdot y(t + \tau)]$$

$$= \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \sin(2\pi t) \cos[2\pi(t + \tau)] dt$$

$$= \frac{1}{2T} \int_{-\frac{T}{2}}^{\frac{T}{2}} 2 \sin(2\pi t) \cos[2\pi(t + \tau)] dt$$

$$= \frac{1}{2T} \int_{-\frac{T}{2}}^{\frac{T}{2}} [\sin(2\pi(2t + \tau)) + \sin 2\pi\tau] dt$$

$$= \frac{1}{2T} \left[\frac{\cos 2\pi(2t + \tau)}{4\pi} \right]_{-\frac{T}{2}}^{\frac{T}{2}} + [\sin 2\pi\tau] \left[\frac{t}{2} \right]_{-\frac{T}{2}}^{\frac{T}{2}}$$

$$= \frac{1}{2T} [(\cos(2\pi T + 2\tau) - \cos(2\pi t + 2\tau)) + \sin 2\pi\tau \cdot T]$$

$$\text{So } R_{xy}(\tau) = \sin 2\pi\tau$$

$$R_{xy}(0) = 0.$$

Choice (A)

22. The triangle is bounded by the region $x = 0, x + y = 1$

$$\text{So } \int_0^1 \int_0^{1-x} c(1-x-y) dx dy = 1$$

$$\Rightarrow \int_0^1 \int_0^{1-x} c(1-x) dx dy - \int_0^1 \int_0^{1-x} cy dx dy = 1$$

$$\Rightarrow \int_0^1 [c(1-x)y]^{1-x} dx - \int_0^1 \left[\frac{cy^2}{2} \right]_0^{1-x} dx = 1$$

$$\Rightarrow \int_0^1 \frac{c(1-x)^2}{2} dx = 1$$

$$\Rightarrow \frac{C}{6} = 1; C = 6. \quad \text{Choice (C)}$$

23. Probability of error in ASK, FSK and PSK have
ASK > FSK > PSK
Matched filter have minimum probability of error is

$$\text{erfc} \sqrt{\frac{E}{N_0}}. \quad \text{Choice (C)}$$

24. Nyquist sampling rate for one signal
 $= 2f_m = 4000$ samples/sec
Now sampling rate $= 1.25 \times 4000$ samples/sec
 $= 5000$ samples/sec
There are total eight time division multiplexed signals
requiring a total of $8(5000) = 40,000$ samples/sec
Now, since each sample is encoded in 8 bits
So resultant bit rate will be

$$\frac{1}{T_b} = 8 \times 40000 = 320 \text{ kb/s}$$

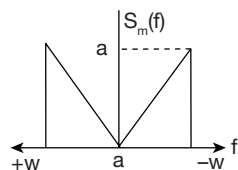
Minimum transmission bond width required is

$$F_B = \frac{1+0.2}{2} (320) = \frac{1.2}{2} \times 320 = 192 \text{ kHz. Choice (D)}$$

25. $H(x) = \sum_{i=1}^n -P_i \log_2 P_i$
 $= \left\{ \frac{1}{4} \log_2 4 + \frac{1}{4} \log_2 4 + \frac{1}{8} \log_2 8 + \frac{3}{8} \log_2 \frac{8}{3} \right\}$
 $= \left\{ \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + 0.53 \right\}$
 $= 1.905$ bits per symbol Choice (A)

$$26. P = \int_{-\infty}^{+\infty} S_m(f) df = 2 \int_0^w \frac{af}{w} df = aW$$

$$(\text{SNR})_0 = \frac{A_c^2 P}{4WN_0}, \text{ here } A_c \text{ is}$$



Carrier amplitude

$$(\text{SNR})_0 = \frac{A_c^2 aW}{4WN_0}$$

$$= \frac{aA_c^2}{4N_0}. \quad \text{Choice (B)}$$

$$27. P_e = Q \left(\frac{\text{Amplitude of binary pulses}}{\text{channel noise rms amp}} \right)$$

$$= Q \left(\frac{1 \times 10^{-3}}{235.8 \times 10^{-6}} \right)$$

$$= Q(4.24) = 1.118 \times 10^{-5}.$$

Choice (C)

28. From the figure

$$f \times (x) = \begin{cases} \frac{1}{9}x - \frac{1}{3} & 3 < x < 6 \\ \frac{-x}{9} + 1 & 6 < x < 9 \end{cases}$$

so for $x = 4.5$

$$f \times (x) = \frac{1}{9}x - \frac{1}{3}$$

$$= \frac{1}{9}(4.5) - \frac{1}{3} = \frac{1}{6}.$$

Choice (A)

29. We know that mean $m_x = \int_{-\infty}^{+\infty} x f_x(x) dx$

$$= \int_{-2}^6 x \frac{1}{8} dx = \frac{1}{8} \int_{-2}^6 x dx$$

$$= \frac{1}{8} \left[\frac{x^2}{2} \right]_{-2}^6 = 2.$$

Variance of a continuous random variable x is expressed as

$$\sigma_x^2 = E[x^2] - m_x^2$$

$$= \int_{-\infty}^{+\infty} x^2 f_x(x) dx - m_x^2$$

$$= \int_{-2}^6 x^2 \cdot \frac{1}{8} dx - m_x^2$$

$$= \frac{1}{8} \left[\frac{x^3}{3} \right]_{-2}^6 - 4 = 5.33.$$

Choice (A)

30. Channel 1 to 30 are multiplexed by first level PCM signals.

So minimum sampling rate required to multiplexed them is

$$f_{s(\min)} = 64 \times 10^3 \times 2 = 128 \text{ kHz}$$

$$\text{So no of frames transmitted/sec} = 128 \times 10^3$$

$$\text{bit rate} \Rightarrow 128 \times 10^3 \times 8 = 2.304 \times 10^6$$

$$\text{So } x = \frac{2.304 \times 10^6}{128 \times 10^3} = 18.$$

Choice (D)

$$31. X(t) = \begin{cases} \frac{2A}{T}t, & 0 < t < T/2 \\ 2A - \frac{2A}{T}t, & T/2 < t < T \end{cases}$$

energy E of the signal $x(t)$ is

$$\begin{aligned}
 E &= \int_{-\infty}^{+\infty} x^2(t) dt = \int_0^{\frac{T}{2}} \left(\frac{2A}{T} t \right)^2 dt + \int_{\frac{T}{2}}^T \left(2A - \frac{2A}{T} t \right)^2 dt \\
 &= \frac{4A^2}{T^2} \left[\frac{t^3}{3} \right]_0^{\frac{T}{2}} + \frac{4A^2 T}{24} \\
 &= \frac{A^2 T}{6} + \frac{A^2 T}{6} = \frac{A^2 T}{3}
 \end{aligned}$$

Maximum signal to noise ratio at output of matched

$$\text{filter is } \left(\frac{S}{N} \right)_{\max} = \frac{2E}{N_0}$$

$$= \frac{2A^2 T}{3N_0} \Rightarrow \frac{2A^2 T}{3N_0}.$$

Choice (B)

32. As we know that

$$\left(\frac{S}{N} \right) = \frac{S_i}{\eta B} = \frac{S_i}{2 \times 10^{-2} \times 10 \times 10^3}$$

$$= 10^4 (= 40 \text{ dB})$$

$$\Rightarrow S_i = 2 \times 10^{-4} \text{ W}$$

Because transmission loss is 70 dB

So required transmission power

$$S_r = (2 \times 10^{-4}) \times 10^7 = 2 \times 10^3 = 2 \text{ KW. Choice (B)}$$

33. For FM with D = 5

$$B_T = 2(D + 1)B$$

$$= 2(6) \times 10 = 120 \text{ KHz.}$$

Choice (B)

34. In binary PCM system

$$L = 2^N$$

$$\left(\frac{S}{N} \right)_{dB} = 1.76 + 20 \log 2^N \quad \text{----- (1)}$$

$$= 1.76 + 6.02 \times N \text{ dB}$$

$$\text{Now since } \left(\frac{S}{N} \right)_{dB} = 30 \text{ dB}$$

$$\left(\frac{S}{N} \right)_{\min} = 1000$$

So from equation (1) we can write

$$20 \log_{10} S = 1.76$$

$$S = 1.22$$

$$\left(\frac{S}{N} \right)_{dB} = 20 \log_{10} 1.22 + 20 \log_{10} L$$

$$10 \log_{10} 10^3 = 20 \log_{10} (L \times 1.22)$$

$$\Rightarrow L = \frac{\sqrt{100}}{1.22} = 25.91 \approx 26 \text{ levels}$$

$$\mu = \log_2 26 = 4.96 \approx 5 \text{ bits}$$

So required of levels are $2^5 = 32$.

Choice (D)

35. Corresponding output signal to quantizing noise ratio will be

$$\left(\frac{S}{N} \right)_{dB} = 1.76 + 6.07 \times 5$$

$$= 31.86 \text{ dB}$$

Choice (B)